

Shiva Garuda

720-925-2427 | shivagaruda23@berkeley.edu | [linkedin.com/in/shiva-garuda](https://www.linkedin.com/in/shiva-garuda) | github.com/Shivag23

EDUCATION

University of California, Berkeley

Berkeley, CA

B.S. in Electrical and Computer Engineering; Expected May 2030

August 2026 – Present

- First-Year Coursework: Linear Algebra & Differential Equations, Structure and Interpretation of Computer Programs, Physics Mechanics & Wave Motion, Introduction to Signals and Systems, Data Structures & Algorithms

RELEVANT EXPERIENCE

1619 Up A Creek Robotics (FRC) – Technical Lead

Aug. 2022 – June 2026

- Captained an 84-member team across software, mechanical, electrical, strategy, and operations, guiding the program to the FIRST Robotics World Championship and earning the **Autonomous Award** for software performance
- Led development of an AprilTag pose-localization subsystem (Java, PhotonVision, ArduCam, Orange Pi coprocessor), reducing localization error by 25%; system won **Innovation in Control Award** at Worlds
- Built 3D simulation tooling in AdvantageScope that cut mechanism design-iteration time by 40%

SciAnno Mosaics (Startup) – Software Engineering Intern

Dec. 2024 – June 2026

- Engineered a Python-based chatbot and agent platform on Google Cloud Platform integrating PubMed, UniProt, and BioProject APIs to automate gene annotation and literature mining, cutting literature-search time by ~20%
- Built 10+ network-inference tools (PPI/co-expression network curation, gene community clustering, adapted Random Walk with Restart), now used by the CU Boulder Kane Lab for leaf transcriptomic research
- Co-authored a paper published in *Nature Scientific Reports* based on contributions to the platform

Kane Lab, CU Boulder – Research Assistant

May 2024 – March 2026

- Built a custom Python data-augmentation pipeline (Augmentor) that expanded a curated cannabis-leaf image dataset to 2,000+ training images
- Landmarked leaf morphology (TPSDig) across monoecious accessions and applied Principal Component Analysis in R to quantify how backcrossing affects *Cannabis sativa* leaf shape
- Trained a CNN on this data achieving 91% accuracy classifying leaf images as monoecious vs. dioecious

PROJECTS

Wearable Posture Monitor | *Arduino/Embedded C, Circuit Design, Sensor Calibration*

July 2025 – May 2026

- Built a wearable posture-correction device with Arduino Nano, buzzer, and battery, achieving a 40% improvement in slouch-detection accuracy and delivering haptic feedback in under 0.5 seconds
- Filed a provisional patent to address rising rates of poor posture among teens from prolonged screen use, and user-tested with 20+ students

Fantasy Football Analyzer | *Python, Node.js, Neural Networks, Vercel, Neon (Postgres)*

July 2026 – Aug. 2026

- Built a full-stack web app that scrapes historical NFL statistics and serves weekly player-performance predictions from a trained neural network
- Designed the front end and deployed on Vercel with a Neon Postgres backend, using multi-agent workflows to accelerate development

Robotic Tic-Tac-Toe Tangible User Interface | *JavaScript, Arduino, RFID, OnShape*

June 2023 – Aug. 2023

- Designed and 3D-printed a custom enclosure in OnShape, then wired and soldered an Arduino, RFID sensor, and servo motor to integrate embedded hardware with Toio robots for autonomous interaction with players
- Programmed robots in JavaScript for autonomous map sweeping and player-controlled TicTacToe
- Mentored 2 teammates to accelerate development and presented HCI Research to 150+ community members

VAE for Neuroimaging Data Generation | *Python, Deep Learning, Google Colab*

May 2024 – Aug. 2024

- Trained a VAE on 3,000+ time-series samples over a 6-week research project at the CU Boulder Imagine AI Lab to generate synthetic neuroimaging data with ~3% reconstruction error; presented findings to lab researchers

SKILLS & TOOLS

Languages: Python, Java, JavaScript, R, HTML/CSS, Embedded C

Frameworks & Platforms: TensorFlow, LabVIEW, Arduino, Node.js, LangChain/LangGraph, Google Cloud Platform, Vercel, PhotonVision, Multi-Agent/Agentic Systems

Hardware & Tools: Circuit Design & Soldering, MPU6050, Orange Pi, ArduCam, RFID, 3D Printing, AdvantageScope, Augmentor, TPSDig, Roboflow, Git, VS Code, Jupyter Notebook, Cursor, Claude Code, Supabase, Neon